



Full Length Article

Spectroscopic properties of Er:Yb:Bi₄Si₃O₁₂ crystal for 1.5–1.6 μm laser

Lincong Rao^{a,c}, Yujin Chen^{a,b}, Jianhua Huang^{a,b}, Xinghong Gong^{a,b}, Yanfu Lin^{a,b}, Zundu Luo^a, Yidong Huang^{a,b,*}

^a Key Laboratory of Optoelectronic Materials Chemistry and Physics, Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences, Fuzhou, Fujian, 350002, China

^b Fujian Science & Technology Innovation Laboratory for Optoelectronic Information of China, Fuzhou, Fujian, 350108, China

^c University of Chinese Academy of Sciences, Beijing 100039, China



ARTICLE INFO

Keywords:

Spectroscopic property
Er:Yb:Bi₄Si₃O₁₂ crystal
1.5–1.6 μm laser

ABSTRACT

An Er³⁺ and Yb³⁺ co-doped Bi₄Si₃O₁₂ single crystal was grown by the Czochralski method and the spectroscopic properties of the crystal were studied at room temperature in detail. The crystal has a peak absorption cross section of $0.98 \times 10^{-20} \text{ cm}^2$ at wavelength of 975 nm, and a peak emission cross section of $1.01 \times 10^{-20} \text{ cm}^2$ at wavelength of 1544 nm. The fluorescence lifetimes of the ⁴I_{11/2} and ⁴I_{13/2} multiplets of Er³⁺ are 14.4 μs and 5.64 ms, respectively. The fluorescence quantum efficiency of the ⁴I_{13/2} multiplet of Er³⁺ and the Yb³⁺ → Er³⁺ energy transfer efficiency are 88.4% and 77.5%, respectively. The results indicate that the Er:Yb:Bi₄Si₃O₁₂ crystal is a promising 1.5–1.6 μm laser gain medium.

1. Introduction

1.5–1.6 μm laser can be widely used in lidar, remote sensing, and optical communication, because of its superiorities of eye-safe as well as locating in the atmospheric transmission window and the lowest attenuation region in silica fiber [1–3]. At present, the commercial gain medium for 1.5–1.6 μm solid-state laser is Er³⁺/Yb³⁺ co-doped phosphate glass (Er:Yb:PG) [1]. However, its poor thermal property limits the output power of 1.5–1.6 μm laser [2]. Due to the good thermal property and high Yb³⁺ → Er³⁺ energy transfer efficiency, Er³⁺/Yb³⁺ co-doped RAl₃(BO₃)₄ (Er:Yb:RAB, R = Lu, Y, and Gd) crystals have been demonstrated as efficient 1.5–1.6 μm laser gain media [4–6]. Continuous-wave (CW) 1.5–1.6 μm lasers with slope efficiency above 20% and maximum output power higher than 1.0 W have been realized in them [4–6]. However, the high phonon energy (>1400 cm⁻¹ [7]) of RAB crystal leads to a strong non-radiative transition from ⁴I_{13/2} to ⁴I_{15/2} multiplets of Er³⁺ and then a short fluorescence lifetime (about 320 μs) of the ⁴I_{13/2} multiplet [8,9], which restricts the energy storage capacity and output energy of Er:Yb:RAB 1.5–1.6 μm pulse laser.

Due to their lower phonon energy (about 1000 cm⁻¹), Er³⁺/Yb³⁺ co-doped silicate crystals have longer fluorescence lifetime (several ms) of the ⁴I_{13/2} multiplet [10–16]. Besides, Yb³⁺ → Er³⁺ energy transfer efficiencies in some Er³⁺/Yb³⁺ co-doped silicate crystals, such as Er:Yb:

Ca₃NbGa₃Si₂O₁₄ (Er:Yb:CNGS), Er:Yb:Ce:Ca₂Al₂SiO₇ (Er:Yb:Ce:CAS), Er:Yb:Lu₂Si₂O₇ (Er:Yb:LPS) and Er:Yb:Y₂SiO₅ (Er:Yb:YSO), are higher than 70% [10–16]. Up to now, a 1565 nm CW laser with an output power of 1.05 W and a slope efficiency of 8.8%, as well as a 1565 nm CW laser with an output power of 2.18 W and a slope efficiency of 21.8% have been realized in the Er:Yb:CNGS and Er:Yb:LPS crystals, respectively [11,14]. Therefore, Er³⁺/Yb³⁺ co-doped silicate crystals have been considered as promising gain media for 1.5–1.6 μm laser.

Bismuth silicate Bi₄Si₃O₁₂ (BSO) crystal belongs to the cubic system with a space group of *I*-43d [17]. The crystal melts congruently at about 1025 °C and can be grown by the Czochralski method [17]. It has a moderate thermal conductivity (3.29 Wm⁻¹K⁻¹ at 298K) and low thermal expansion coefficient (7.07 × 10⁻⁶ K⁻¹) [18]. Therefore, the isotropic BSO crystals doped with some rare earth ions have been investigated as laser gain media [19–21]. For example, a 1040 nm CW laser with an output power of 213 mW and a slope efficiency of 57.6%, as well as a 1909 nm CW laser with an output power of 650 mW and a slope efficiency of 29.7% have been obtained in the Yb:BSO and Tm:BSO crystals, respectively [19,20]. Furthermore, some spectroscopic properties of Er³⁺ singly doped BSO crystal, including the absorption and emission spectra as well as J-O theory analysis, have also been studied [22]. However, the spectroscopic properties of Er³⁺/Yb³⁺ co-doped BSO (Er:Yb:BSO) crystal have not been reported up to now, such as the

* Corresponding author. Key Laboratory of Optoelectronic Materials Chemistry and Physics, Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences, Fuzhou, Fujian, 350002, China.

E-mail address: huyd@fjirsm.ac.cn (Y. Huang).

<https://doi.org/10.1016/j.jlumin.2022.119479>

Received 14 September 2022; Received in revised form 22 October 2022; Accepted 28 October 2022

Available online 1 November 2022

0022-2313/© 2022 Elsevier B.V. All rights reserved.

fluorescence lifetimes of the $^4I_{13/2}$ and $^4I_{11/2}$ multiplets, $Yb^{3+} \rightarrow Er^{3+}$ energy transfer efficiency, as well as up-conversion fluorescence. In this work, an Er:Yb:BSO crystal was grown by the Czochralski method, and its spectroscopic properties related to the 1.5–1.6 μm laser were studied in detail.

2. Experiment

An Er:Yb:BSO single crystal was grown by the Czochralski method in N_2 atmosphere with an iridium crucible. The raw material of Er:Yb:BSO was synthesized by the solid-state reaction method. The concentrations of Er^{3+} and Yb^{3+} in the raw material were 0.6 and 5.0 at.%, respectively. During the crystal growth, the rotating and pulling rates were 10–15 rpm and 0.4–1.0 mm/h, respectively. The concentrations of Er^{3+} and Yb^{3+} in the as-grown crystal were measured by the inductively coupled plasma atomic emission spectrometry (ICP-AES, Ultima2, Jobin-Yvon). The powder X-ray diffraction (XRD) analysis was performed on a diffractometer (Rigaku, Miniflex-600) equipped with $CuK\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$). Room-temperature (RT) absorption spectrum in 330–1700 nm for the Er:Yb:BSO crystal was recorded by an UV/VIS/NIR spectrophotometer (Lambda-950, PerkinElmer). Excited by a xenon lamp at 940 nm, RT fluorescence spectrum in 950–1675 nm was recorded by a spectrometer (FLS1000, Edinburgh). In order to reduce the deformation of fluorescence spectrum caused by the radiation trapping effect, the exciting light was focused on the edge of crystal wafer [23]. The fluorescence decay curves at 980, 1050 and 1544 nm were recorded by a spectrometer (FLS980, Edinburgh), when the exciting wavelengths were 520, 950 and 980 nm, respectively. Excited by a 980 nm diode laser, RT up-conversion fluorescence spectrum in 500–700 nm was recorded by the spectrometer (FLS1000, Edinburgh).

3. Results and discussion

3.1. Crystal characterization

Because the BSO crystal belongs to the cubic system, a $2.5 \times 4.4 \times 5.1 \text{ mm}^3$ transparent crystal wafer without orientation was cut from the as-grown crystal, as shown in Fig. 1(a), and used for the spectral measurements. The concentrations of Er^{3+} and Yb^{3+} in the as-grown crystal were measured to be $1.14 \times 10^{20} \text{ cm}^{-3}$ (0.78 at.%) and $7.45 \times 10^{20} \text{ cm}^{-3}$ (5.09 at.%), respectively. The XRD pattern in $15\text{--}60^\circ$ of the as-grown crystal is presented in Fig. 1(b). All the diffraction peak positions are basically consistent with those of the BSO (PDF # 35–1007), which means that a pure phase Er:Yb:BSO crystal is obtained. However, as can be seen from the magnified XRD pattern in $27\text{--}28^\circ$, the diffraction peak positions of the Er:Yb:BSO crystal shift slightly to a higher angle

than those of the pure BSO, which may be caused by the smaller ionic radii of Er^{3+} (0.89 $\text{\AA}$) and Yb^{3+} (0.87 $\text{\AA}$) than that of Bi^{3+} (1.03 $\text{\AA}$).

3.2. Absorption spectrum and Judd-Ofelt analysis

RT absorption cross section spectrum of the Er:Yb:BSO crystal in 330–1700 nm is presented in Fig. 2. Based on Judd-Ofelt (J-O) theory [24,25], the experimental oscillator strength f_{exp} and the calculated oscillator strength f_{cal} , as well as intensity parameters Ω_t ($t = 2, 4, 6$) of the crystal were calculated and are listed in Table 1. Then, using the calculated J-O intensity parameters Ω_t , the spontaneous transition probability A , fluorescence branching ratio β , and radiative lifetime τ_r for different fluorescence multiplets of Er^{3+} in the crystal were also calculated, and are listed in Table 2. The calculation procedure is similar to that reported in Ref. [22].

The absorption band between 875 and 1050 nm, originated from the transitions of $^4I_{15/2} \rightarrow ^4I_{11/2}$ of Er^{3+} and $^2F_{7/2} \rightarrow ^2F_{5/2}$ of Yb^{3+} , was not taken into account during the calculation of the spectroscopic parameters. The peak absorption cross section is $0.98 \times 10^{-20} \text{ cm}^2$ at wavelength of 975 nm, which is the same as that ($0.98 \times 10^{-20} \text{ cm}^2$) reported for the Yb:BSO crystal [20]. The full width at half the maximum (FWHM) of this absorption band is 2.8 nm.

3.3. Emission and gain spectra

RT fluorescence spectrum in 950–1675 nm was corrected based on the detecting sensitivity of the used spectrometer at various wavelengths, and is presented in Fig. 3. The fluorescence intensity in 1450–1675 nm originating from the $^4I_{13/2} \rightarrow ^4I_{15/2}$ transition of Er^{3+} is much higher than that in 950–1100 nm corresponding to the transitions of $^2F_{5/2} \rightarrow ^2F_{7/2}$ of Yb^{3+} and $^4I_{11/2} \rightarrow ^4I_{15/2}$ of Er^{3+} . It manifests that most of the excited Er^{3+} ions can be populated at the $^4I_{13/2}$ multiplet by efficient resonant energy transfer from the $^2F_{5/2}$ multiplet of Yb^{3+} to the $^4I_{11/2}$ multiplet of Er^{3+} and non-radiative relaxation from $^4I_{11/2}$ to $^4I_{13/2}$ multiplet for Er^{3+} .

Based on the measured RT fluorescence spectrum in 1450–1675 nm, the stimulated emission cross section σ_{em} at wavelength λ can be calculated by the Fuchtbauer-Ladenburg formula [26]:

$$\sigma_{em}(\lambda) = \frac{\lambda^5}{8\tau_r\pi c n^2} \frac{I(\lambda)}{\int \lambda I(\lambda) d\lambda}$$

where τ_r is the radiative lifetime (6.38 ms calculated by the J-O theory), c is the speed of light, n is the refractive index (1.955 at 1544 nm [27]), $I(\lambda)$ is the fluorescence intensity at λ . The stimulated emission cross section spectrum in 1450–1675 nm was calculated and is presented in

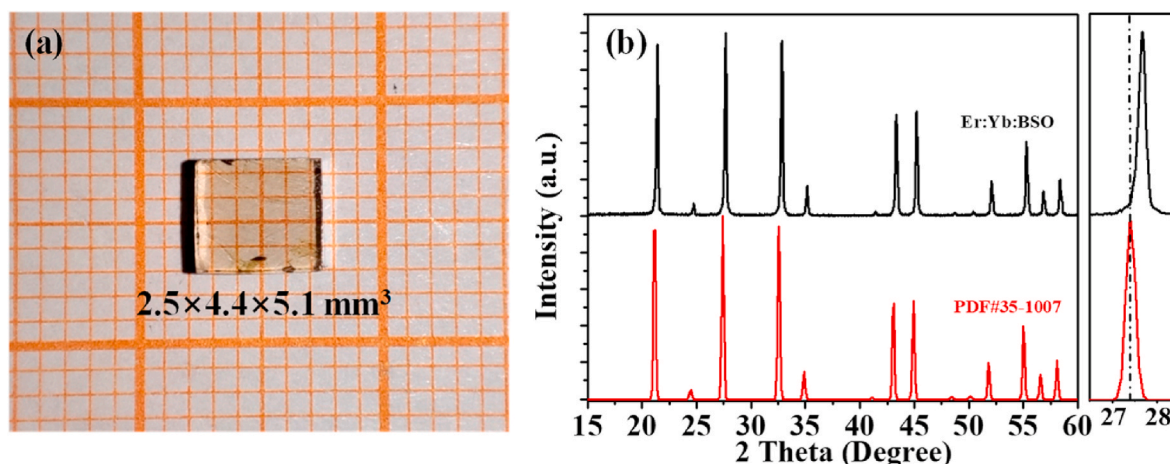


Fig. 1. Photograph (a) and powder XRD pattern (b) of the Er:Yb:BSO crystal.

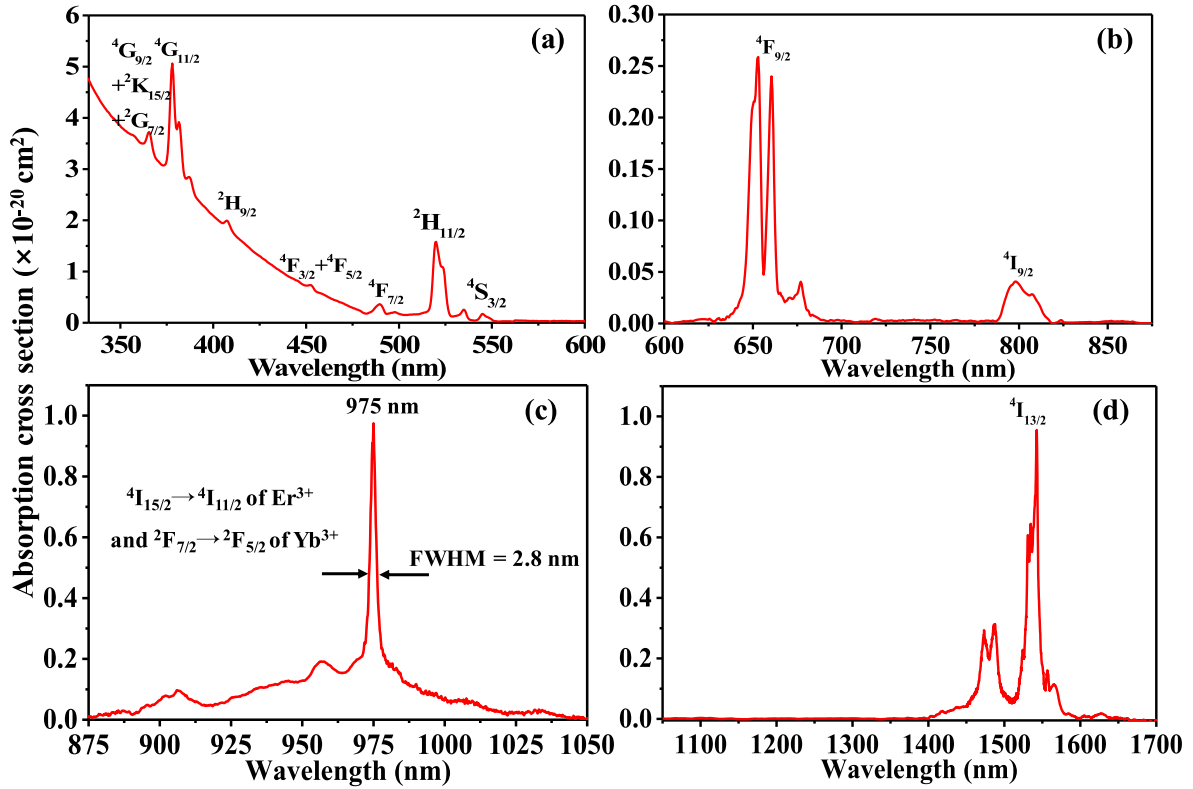


Fig. 2. RT absorption cross section spectrum in 330–1700 nm for the Er:Yb:BSO crystal.

Table 1

Experimental and calculated oscillator strengths for different transitions as well as J-O intensity parameters of Er^{3+} in the Er:Yb:BSO crystal.

Transition (from $4I_{15/2}$)	$\bar{\lambda}$ (nm)	f_{exp} (10^{-6})	f_{cal} (10^{-6})
$4I_{13/2}$	1523	0.783 (ed)	0.714 (ed)
		0.603 (md)	0.603 (md)
$4I_{9/2}$	798	0.127	0.093
$4F_{9/2}$	656	0.875	0.846
$4S_{3/2}$	546	0.297	0.311
$2H_{11/2}$	523	4.605	4.651
$4F_{7/2}$	489	0.751	1.145
$4F_{5/2} + 4F_{3/2}$	453	0.283	0.634
$2H_{9/2}$	408	0.432	0.496
$4G_{11/2}$	381	9.529	9.481
RMS Δf (10^{-6})	0.22		
RMS error	6.18%		
Intensity parameter (10^{-20} cm^2)	$\Omega_2 = 2.088, \Omega_4 = 0.286, \Omega_6 = 0.29$		

Fig. 4(a). The peak emission cross section is $1.01 \times 10^{-20} \text{ cm}^2$ at wavelength of 1544 nm, which is close to that ($1.13 \times 10^{-20} \text{ cm}^2$ [18]) of the reported Er:BSO crystal. The FWHM of the emission band is 15.6 nm.

For predicting the laser oscillating wavelength of the Er:Yb:BSO crystal, the gain cross section spectrum σ_g in 1500–1675 nm can be calculated by Ref. [16]:

$$\sigma_g(\lambda) = \beta \sigma_{em}(\lambda) - (1 - \beta) \sigma_{abs}(\lambda)$$

where β is the ratio of the number of Er^{3+} ions in the upper laser multiplet $4I_{13/2}$ to the total number of Er^{3+} ions. $\sigma_{em}(\lambda)$ and $\sigma_{abs}(\lambda)$ are the emission and absorption cross sections at wavelength λ , respectively, which are presented in Fig. 2(d). It can be seen from the gain curves shown in Fig. 4(b) that the σ_g turns into a positive value around 1638 nm when $\beta = 0.1$. When $\beta = 0.4$, peak gains can be observed at wavelengths of 1544, 1559, 1569 and 1638 nm. Therefore, multi-wavelength laser oscillation may be expected in the Er:Yb:BSO crystal, which may be

Table 2

Spontaneous transition probability, fluorescence branching ratio, and radiative lifetime of Er^{3+} in the Er:Yb:BSO crystal.

$J \rightarrow J'$	$\bar{\lambda}$ (nm)	A (s^{-1})	β (%)	τ_r (ms)
$4I_{13/2} \rightarrow$				6.381
$4I_{15/2}$	1549	156.712	100	
$4I_{11/2} \rightarrow$				7.951
$4I_{13/2}$	3031	10.794	8.58	
$4I_{15/2}$	1025	114.972	91.42	
$4I_{9/2} \rightarrow$				9.316
$4I_{11/2}$	3686	1.547	1.44	
$4I_{13/2}$	1663	44.113	41.09	
$4I_{15/2}$	802	61.687	57.47	
$4F_{9/2} \rightarrow$				1.071
$4I_{9/2}$	3879	2.521	0.27	
$4I_{11/2}$	1890	64.751	6.93	
$4I_{13/2}$	1164	35.516	3.8	
$4I_{15/2}$	665	831.158	89	
$4S_{3/2} \rightarrow$				0.604
$4F_{9/2}$	3223	0.546	0.03	
$4I_{9/2}$	1761	38.054	2.3	
$4I_{11/2}$	1192	32.401	1.96	
$4I_{13/2}$	855	418.555	25.27	
$4I_{15/2}$	551	1166.8	70.44	
$2H_{11/2} \rightarrow$				0.151
$4S_{3/2}$	13,063	0.012	0	
$4F_{9/2}$	2585	19.274	0.292	
$4I_{9/2}$	1551	71.751	1.09	
$4I_{11/2}$	1092	47.043	0.712	
$4I_{13/2}$	803	87.291	1.32	
$4I_{15/2}$	529	6378.5	96.586	

further used to generate the THz radiation by the difference frequency technology between these wavelengths [28].

3.4. Fluorescence lifetimes

Excited by an OPO laser at 520 nm, the fluorescence decay curve at

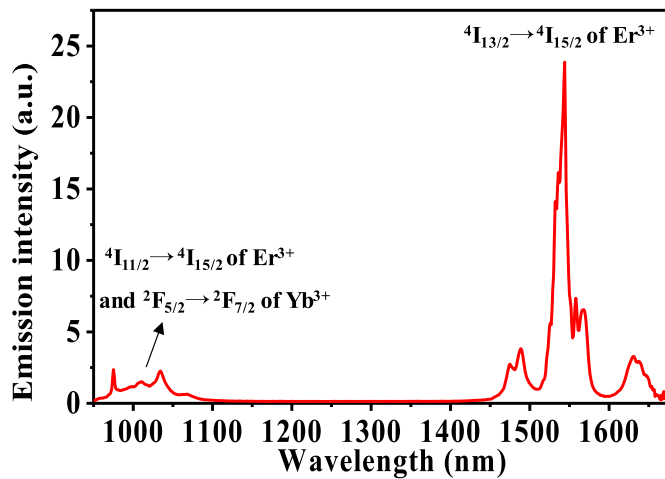


Fig. 3. RT fluorescence spectrum for the Er:Yb:BSO crystal in 950–1675 nm under excitation at 940 nm.

980 nm, corresponding to the $^4I_{11/2}$ multiplet of Er^{3+} , was measured from the (0.6 at.%) Er:BSO crystalline powder prepared by the high temperature solid-state method. Excited by a xenon lamp at 975 nm, the fluorescence decay curve at 1544 nm, corresponding to the $^4I_{13/2}$ multiplet of Er^{3+} , was measured from the Er:Yb:BSO crystal powder. For reducing the radiation trapping effect, the crystal power was immersed

in the chlorobenzene ($n = 1.52$) [29]. As presented in Fig. 5, the fluorescence decays at 980 and 1544 nm both exhibit simple exponential behavior. The fluorescence lifetime of $^4I_{11/2}$ multiplet of Er^{3+} in the BSO crystal was fitted to be 14.4 μs . The fluorescence lifetime of the $^4I_{13/2}$ multiplet of Er^{3+} in the BSO crystal was fitted to be 7.3 ms, and is still longer than the calculated radiative lifetime (6.38 ms), which may be caused by the radiation trapping effect remaining in the powder sample [29].

In order to avoid the radiation trapping effect and obtain the intrinsic fluorescence lifetime of the $^4I_{13/2}$ multiplet, the pinhole method was also adopted in the measurement of fluorescence decay curve [30]. Excited by a 980 nm pulse laser, the decay curves at 1544 nm for different pinhole diameter ϕ were measured and the experimental results are presented in Fig. 6(a). It can be seen from Fig. 6(b) that the fluorescence lifetime increases linearly with the pinhole diameter. Theoretically, when the pinhole diameter is reduced to zero, the influence of the radiation trapping effect on the measured fluorescence lifetime can be avoided [30]. By linear fitting and extrapolating, the intrinsic fluorescence lifetime of the $^4I_{13/2}$ multiplet was estimated to be 5.64 ms, which is shorter than the calculated radiative lifetime (6.38 ms). Then, the fluorescence quantum efficiency of the $^4I_{13/2}$ multiplet was calculated to be 88.4%, which is far higher than that (7%) of the Er:Yb:YAB crystal [5] and close to that (about 90%) of the Er:Yb:PG [31].

3.5. Energy transfer

The energy transfer efficiency η_{ET} from the $^2F_{5/2}$ multiplet of Yb^{3+} to

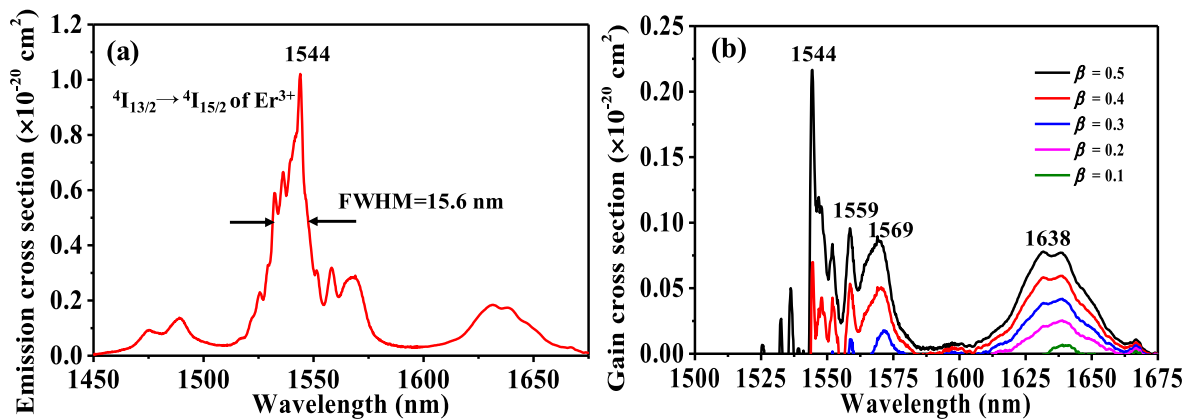


Fig. 4. RT emission cross section spectrum in 1450–1675 nm (a) and gain cross section spectra in 1500–1675 nm (b) for the Er:Yb:BSO crystal.

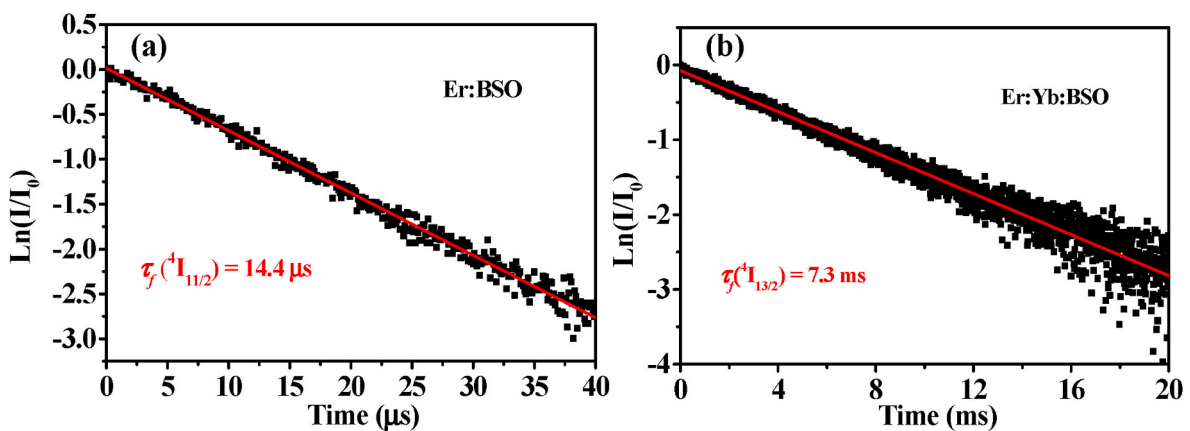


Fig. 5. Fluorescence decay curve at 980 nm, corresponding to the transition of $^4I_{11/2} \rightarrow ^4I_{15/2}$ of Er^{3+} , for the Er:BSO crystalline powder under excitation at 520 nm (a); and fluorescence decay curve at 1544 nm, corresponding to the transition of $^4I_{13/2} \rightarrow ^4I_{15/2}$ of Er^{3+} , for the Er:Yb:BSO crystal powder under excitation at 980 nm (b).

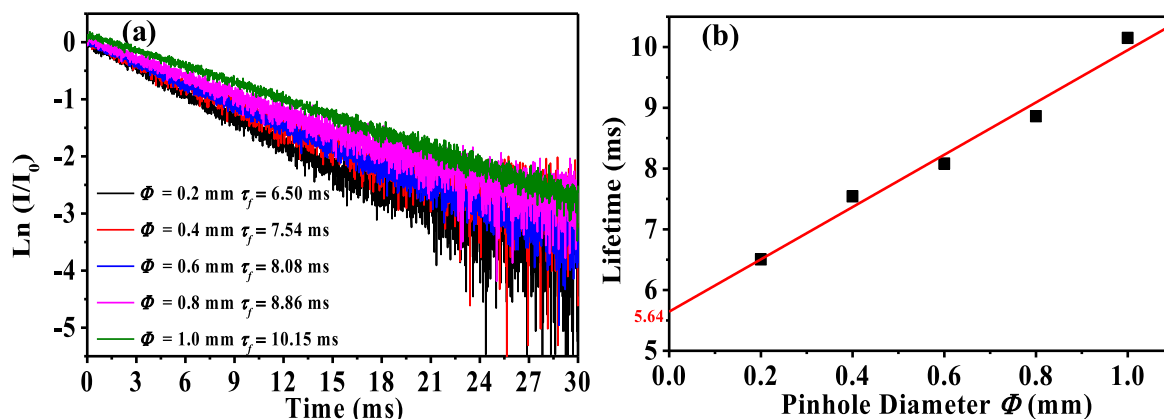


Fig. 6. Fluorescence decay curves at 1544 nm, corresponding to the transition of $^4I_{13/2} \rightarrow ^4I_{15/2}$ of Er^{3+} , under excitation at 980 nm for different pinhole diameters ϕ (a) and extrapolation of the measured lifetimes (b) of the Er:Yb:BSO crystal.

the $^4I_{11/2}$ multiplet of Er^{3+} in the Er:Yb:BSO crystal can be calculated by Ref. [32]:

$$\eta_{ET} = 1 - \tau_{\text{co-doped}} / \tau_{\text{singly-doped}}$$

where $\tau_{\text{co-doped}}$ and $\tau_{\text{singly-doped}}$ are the fluorescence lifetimes of the $^2F_{5/2}$ multiplet of Yb^{3+} in the Er:Yb:BSO and Yb:BSO crystals, respectively. Excited by a xenon lamp at 950 nm, the fluorescence decay curves at 1050 nm were recorded from the Er:Yb:BSO crystal powder and (5 at.%) Yb:BSO crystalline powder prepared by the high temperature solid-state method. The two powers were also immersed in the chlorobenzene ($n = 1.52$) to reduce the radiation trapping effect. As shown in Fig. 7, two decay curves also exhibit the simple exponential behavior. The fluorescence lifetimes of the $^2F_{5/2}$ multiplet of Yb^{3+} in the Er:Yb:BSO and Yb:BSO crystals were fitted to be 247 μs and 1.1 ms, respectively. Then, the η_{ET} in the Er:Yb:BSO crystal was calculated to be 77.5%.

3.6. Up-conversion fluorescence

RT up-conversion fluorescence spectrum of the crystal in 500–700 nm is presented in Fig. 8(a). It can be seen that the spectrum is dominated by the green fluorescence band centered at 546 nm corresponding to the $^4S_{3/2} \rightarrow ^4I_{15/2}$ transition. Two weak fluorescence bands centered at 525 and 655 nm, corresponding to the $^2H_{11/2} \rightarrow ^4I_{15/2}$ and $^4F_{9/2} \rightarrow ^4I_{15/2}$ transitions, respectively, were also observed. The up-conversion fluorescence can be explained based on the energy level diagram of $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped material shown in Fig. 8(b). After Yb^{3+} ions absorb the exciting energy at 980 nm, Er^{3+} ions can be populated at the $^4I_{11/2}$

multiplet via resonant energy transfer (ET1). Then a part of Er^{3+} ions at the $^4I_{11/2}$ multiplet can populate the $^4F_{7/2}$ multiplet via excited state absorption (ESA1) and energy transfer (ET2). Through the multiphonon non-radiative relaxation, Er^{3+} ions at the $^4F_{7/2}$ multiplet can populate the $^2H_{11/2}$, $^4S_{3/2}$ and $^4F_{9/2}$ multiplets and then transit to the $^4I_{15/2}$ multiplet, emitting the fluorescence bands centered at 525, 546 and 655 nm, respectively. Another part of Er^{3+} ions at the $^4I_{11/2}$ multiplet can firstly relax to the $^4I_{13/2}$ multiplet via multiphonon non-radiative transition, and then transit to the $^4F_{9/2}$ multiplet via excited state absorption (ESA2) and energy transfer (ET3). Finally, Er^{3+} ions at the $^4F_{9/2}$ multiplet transition to the $^4I_{15/2}$ multiplet, emitting the red fluorescence band centered at 655 nm. The up-conversion fluorescence will reduce the number of Er^{3+} ions at the $^4I_{13/2}$ upper laser multiplet and decrease the slope efficiency of 1.55 μm laser.

Some spectroscopic and thermo-mechanical parameters of the $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped silicate crystals and Er:Yb:PG are listed in Table 3. The peak absorption cross section ($0.98 \times 10^{-20} \text{ cm}^2$) at 975 nm of the Er:Yb:BSO crystal is close to that ($1.0 \times 10^{-20} \text{ cm}^2$) at 976 nm of the Er:Yb:PG, but smaller than those at 978 nm of the other $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped silicate crystals listed in the table. The peak emission cross section ($1.01 \times 10^{-20} \text{ cm}^2$) at 1544 nm of the Er:Yb:BSO crystal is close to those of the listed $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped materials. Compared with those of the listed $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped materials, the smaller FWHMs of the absorption and fluorescence bands of the Er:Yb:BSO crystal may be caused by the ordered structure of the BSO crystal. In the Er:Yb:BSO crystal, both Er^{3+} and Yb^{3+} ions occupy the Bi^{3+} sites with point group symmetry C_3 . The narrow absorption band means that the emission wavelength of the used

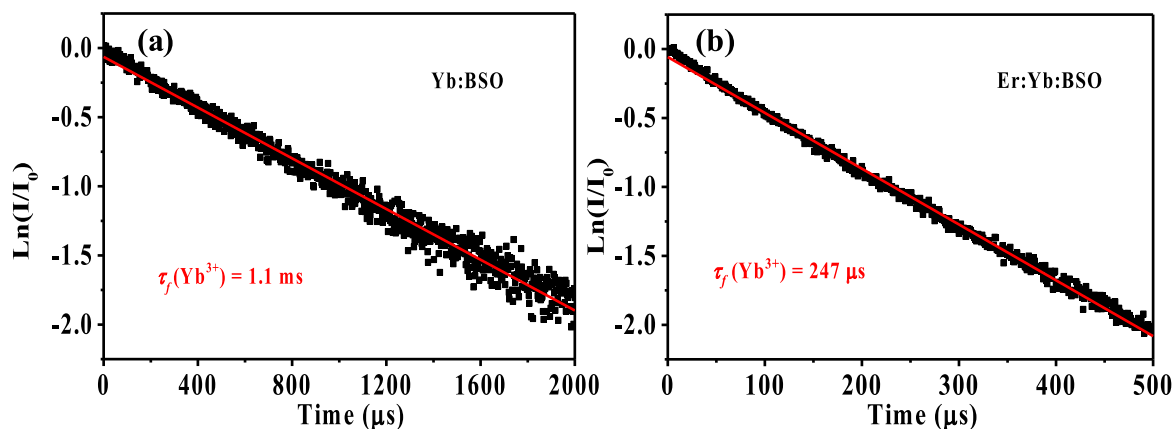


Fig. 7. Fluorescence decay curves in the Yb:BSO crystalline powder (a) and Er:Yb:BSO crystal (b) at 1050 nm, corresponding to the transition of $^2F_{5/2} \rightarrow ^2F_{7/2}$ of Yb^{3+} , under excitation at 950 nm.

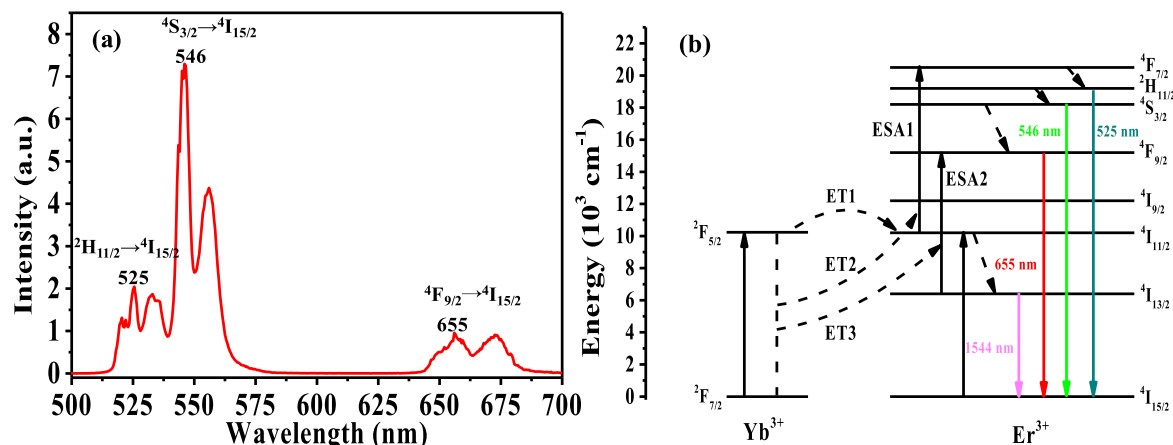


Fig. 8. RT up-conversion fluorescence spectrum in 500–700 nm of the Er:Yb:BSO crystal excited at 980 nm (a) and Energy level diagram of Er³⁺/Yb³⁺ co-doped material (b).

Table 3

Some spectroscopic and thermo-mechanical parameters of the Er³⁺/Yb³⁺ co-doped phosphate glass and silicate crystals.

Material	Er:Yb:BSO	Er:Yb:PG	Er:Yb:CNGS	Er:Yb:Ce:CAS	Er:Yb:LPS	Er:Yb:YSO
Crystal system	Cubic	glass	trigonal	trigonal	monoclinic	monoclinic
σ_{abs} (10^{-20} cm ²)	0.98@975	1.0@976	3.60 (σ)	2.75 (σ)	1.57 (E//Z) @978	2.3 ($\parallel$ X) @978
@ λ_p (nm)			@978	@978		
FWHM of absorption band (nm)	2.8	6	4.6 (σ)	4.6 (σ)	7.2 ($\parallel$ Z)	3.5 ($\parallel$ X)
σ_{em} (10^{-20} cm ²)	1.01@1544	0.8@1533	0.84 (σ)		0.81 ($\parallel$ Z) @1537	0.9 ($\parallel$ X) @1540
@ λ_e (nm)			@1533			
FWHM of emission band (nm)	15.6	45	60 (α)	30 (σ)		28 ($\parallel$ X)
τ_f of ⁴ I _{11/2} multiplet (μ s)	14.4	2–3	29.7	41	8.33	16
τ_f of ⁴ I _{13/2} multiplet (ms)	5.64	8	5.81	8	8.68	8
Energy transfer efficiency (%)	77.5	90	72	78	85	75
Melting point (°C)	1025		1345	1580	1900	2000
Thermal conductivity (Wm ⁻¹ K ⁻¹)	3.29 (pure)	0.8	1.82 ($\parallel$ c)	2.2 (pure, $\parallel$ c)	7.29 ($\parallel$ Z)	4.4 (pure)
References	This work [24],	[1,2,29,33]	[10,34]	[12,35]	[13,14]	[15,16,36]

LD pumping source needs to be controlled more accurately. The energy transfer efficiency (77.5%) of the Er:Yb:BSO crystal is lower than those of the Er:Yb:PG (90%) and Er:Yb:LPS (85%) crystal, but close to those (about 75%) of other Er³⁺/Yb³⁺ co-doped silicate crystals. By further optimizing the concentrations of Er³⁺ and Yb³⁺, the energy transfer efficiency in the Er:Yb:BSO crystal may be improved in the future. Furthermore, the Er:Yb:BSO crystal has a lower melting point (only 1025 °C), but a higher thermal conductivity than the Er:Yb:CNGS and Er:Yb:Ce:CAS crystals. Considering the isotropic characteristic of the thermal conductivity and expansion coefficient, it can be inferred that the Er:Yb:BSO crystal is more suitable for laser gain medium. Although the Er:Yb:LPS and Er:Yb:YSO crystals have higher thermal conductivities, the high melting point (≥ 1900 °C) will cause the crystals grown with high cost and difficulty.

4. Conclusion

An Er:Yb:BSO crystal was grown by the Czochralski method. Its spectroscopic properties are comparable to those of Er³⁺/Yb³⁺ co-doped silicate crystals in which efficient laser operations have been realized. Combined with its moderate thermal conductivity and low thermal expansion coefficient, the Er:Yb:BSO crystal may be a promising candidate for eye-safe laser at 1.5–1.6 μ m.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant 52272010), National Key Research and Development Program of China (2021YFB3601504), Fujian Science & Technology Innovation Laboratory for Optoelectronic Information of China (2021ZR119, 2021ZZ118) and Scientific Instrument Developing Project of the Chinese Academy of Sciences (YZLY202001).

References

- [1] B. Denker, B. Galagan, S. Sverchkov, A. Prokhorov, Erbium (Er) glass lasers, in: B. Denker, E. Shklovsky (Eds.), Handbook of Solid-State Lasers, Woodhead, 2013, pp. 341–358.
- [2] P. Laporta, S. Taccheo, S. Longhi, O. Svelto, C. Svelto, Erbium–ytterbium microlasers: optical properties and lasing characteristics, Opt. Mater. 11 (1999) 269–288.
- [3] P. Lutzmann, B. Göhler, F.A. Puttena, C.A. Hillb, Laser Vibration Sensing: Overview and Applications, 2011, pp. 1–16. SPIE 818602.
- [4] Y.J. Chen, Y.F. Lin, J.H. Huang, X.H. Gong, Z.D. Luo, Y.D. Huang, Spectroscopic and laser properties of Er³⁺:Yb³⁺:LuAl₃(BO₃)₄ crystal at 1.5–1.6 μ m, Opt Express 18 (13) (2010) 13700–13707.
- [5] N.A. Tolstik, S.V. Kurilchik, V.E. Kisel, N.V. Kuleshov, V.V. Maltsev, O.V. Pilipenko, E.V. Kopyulina, N.I. Leonyuk, Efficient 1 W continuous-wave diode-pumped Er:Yb:YAl₃(BO₃)₄ laser, Opt. Lett. 32 (22) (2007) 3233–3235.
- [6] K.N. Gorbachenya, V.E. Kisel, A.S. Yasukevich, V.V. Maltsev, N.I. Leonyuk, N. V. Kuleshov, Eye-safe 1.55 μ m passively Q-switched Er:Yb:GdAl₃(BO₃)₄ diode-pumped laser, Opt. Lett. 41 (5) (2016) 918–921.
- [7] A.S. Oreshonkov, E.M. Roginskii, N.P. Shestakov, I.A. Gudim, V.L. Temerov, I. V. Nemtsev, M.S. Molokeev, S.V. Adichtchev, A.M. Pugachev, Y.G. Denisenko,

- Structural, electronic and vibrational properties of $\text{YAl}_3(\text{BO}_3)_4$, *Materials* 13 (545) (2020) 1–16.
- [8] N.A. Tolstik, V.E. Kisel, N.V. Kuleshov, V.V. Maltsev, N.I. Leonyuk, Yb Er, $\text{YAl}_3(\text{BO}_3)_4$ -efficient 1.5 μm laser crystal, *Appl. Phys. B* 97 (2009) 357–362.
 - [9] Y.J. Chen, Y.F. Lin, X.H. Gong, Z.D. Luo, Y.D. Huang, Spectroscopic properties and laser performance of Er^{3+} and Yb^{3+} co-doped $\text{GdAl}_3(\text{BO}_3)_4$ crystal, *IEEE J. Quant. Electron.* 43 (10) (2007) 950–956.
 - [10] Y.J. Chen, G.L. Gong, J.H. Huang, Y.F. Lin, X.H. Gong, Z.D. Luo, Y.D. Huang, Spectra and diode-pumped continuous-wave 1.55 μm laser of $\text{Er}:\text{Yb}:\text{Ca}_3\text{NbGa}_3\text{Si}_2\text{O}_{14}$ crystal, *Opt Express* 27 (24) (2019) 35418–35425.
 - [11] Y.J. Chen, J.H. Huang, Y.F. Lin, X.H. Gong, Z.D. Luo, Y.D. Huang, Continuous-wave and passively Q-switched pulse $\text{Er}:\text{Yb}:\text{Ca}_3\text{NbGa}_3\text{Si}_2\text{O}_{14}$ microlasers at 1.5–1.6 μm , *Opt. Mater.* 131 (2022), 112733.
 - [12] A. Jaffrès, P. Loiseau, G. Aka, B. Viana, C. Larat, E. Lallier, CW diode pumped Er, Yb,Ce:CaS single crystal 1.5 μm laser, *Laser Phys.* 24 (2014), 125801.
 - [13] J.H. Huang, Y.J. Chen, H. Wang, Y.F. Lin, X.H. Gong, Z.D. Luo, Y.D. Huang, Efficient 1620 nm continuous-wave laser operation of Czochralski grown $\text{Er}:\text{Yb}:\text{Lu}_2\text{Si}_2\text{O}_7$ crystal, *Opt Express* 25 (20) (2017) 24001–24006.
 - [14] Y.H. Chen, J.H. Huang, Y.D. Huang, X.H. Gong, Y.F. Lin, Z.D. Luo, Y.J. Chen, Refractive index, thermal, spectroscopic and 1.55 μm laser properties of an $\text{Er}:\text{Yb}:\text{Lu}_2\text{Si}_2\text{O}_7$ crystal, *Opt. Mater.* 128 (2022), 112448.
 - [15] T. Schweizer, T. Jensen, E. Heumann, G. Huber, Spectroscopic properties and diode-pumped 1.6 μm laser performance in Yb-codoped $\text{Er}:\text{Y}_3\text{Al}_5\text{O}_{12}$, and $\text{Er}:\text{Y}_2\text{SiO}_5$, *Opt Commun.* 118 (5) (1995) 557–561.
 - [16] C. Li, R. Moncorge, J.C. Souriau, C. Borel, C. Wyon, Room temperature cw laser action of $\text{Y}_2\text{SiO}_5:\text{Yb}^{3+}$, Er^{3+} at 1.57 μm , *Opt Commun.* 107 (1994) 61–64.
 - [17] Hua Jiang, H.J. Kim, G. Rooh, H. Park, S. Kim, J.K. Cheon, Czochralski growth and scintillation properties of $\text{Bi}_4\text{Si}_3\text{O}_{12}$ (BSO) single crystal, *Nucl. Instrum. Methods Phys. Res.* 648 (2011) 73–76.
 - [18] H. Shen, J.Y. Xu, W.J. Ping, Q.B. He, Y. Zhang, M. Jin, G.J. Jiang, Growth, mechanical and thermal properties of $\text{Bi}_4\text{Si}_3\text{O}_{12}$ single crystals, *Chin. Phys. Lett.* 29 (7) (2012), 0765.
 - [19] H. Lin, Z. Lin, Y. Chen, Y. Huang, G. Zhang, W. Chen, Continuous-wave and SESAM mode-locked operation of the Yb: $\text{Bi}_4\text{Si}_3\text{O}_{12}$ laser, *Opt Express* 29 (1/4) (2021) 105–110.
 - [20] Y. Zhang, L. Huang, X. Duan, Z. Gong, T. Tian, Y. Ma, J. Xu, $\text{Tm}^{3+}:\text{Bi}_4\text{Si}_3\text{O}_{12}$ crystal as a promising laser material near 2 μm : growth, spectroscopic properties and laser performance, *Opt Express* 29 (18) (2021) 29138–29148.
 - [21] Y. Zhang, Z. Gong, X.M. Duan, J.P. Yuan, J.Y. Xu, Y.F. Ma, Y.J. Shen, Growth, spectra and continuous-wave 2.1 μm laser operation of a Ho^{3+} -doped bismuth silicate crystal, *CrystEngComm* 24 (2022) 1590–1597.
 - [22] A. Lira, I. Camarillo, E. Camarillo, F. Ramos, M. Flores, U. Caldino, Spectroscopic characterization of Er^{3+} transitions in $\text{Bi}_4\text{Si}_3\text{O}_{12}$, *J. Phys. Condens. Matter* 16 (2004) 5925–5936.
 - [23] A.B. Suchocki, G. D. Gilliland, R. C. Powell, J. M. Bowen, J.C. Walling, Spectroscopic properties of alexandrite crystals II, *J. Lumin.* 37 (1987) 29–37.
 - [24] B. Judd, Optical absorption intensities of rare-earth ions, *Phys. Rev.* 127 (3) (1962) 750–761.
 - [25] G. Ofelt, Intensities of crystal spectra of rare earth ions, *J. Chem. Phys.* 37 (3) (1962) 511–520.
 - [26] B. Aull, H. Jenssen, Vibronic interactions in Nd: YAG resulting in nonreciprocity of absorption and stimulated emission cross sections, *IEEE J. Quant. Electron.* 18 (1982) 925–930.
 - [27] D. Bortfeld, H. Meier, Refractive indices and electro-optic coefficients of the eulites $\text{Bi}_4\text{Ge}_3\text{O}_{12}$ and $\text{Bi}_4\text{Si}_3\text{O}_{12}$, *J. Appl. Phys.* 43 (12) (1972) 5110–5111.
 - [28] M.R.K. Soltanian, I.S. Amiri, S.E. Alavi, H. Ahmad, Dual-wavelength erbium-doped fiber laser to generate terahertz radiation using photonic crystal fiber, *J. Lightwave Technol.* 33 (24) (2015) 5038–5046.
 - [29] D.S. Sumida, T.Y. Fan, Effect of radiation trapping on fluorescence lifetime and emission cross section measurements in solid-state laser media, *Opt. Lett.* 19 (17) (1994) 1343–1345.
 - [30] H. Kühn, S.T. Fredrich-Thornton, C. Kränkel, R. Peters, K. Petermann, Model for the calculation of radiation trapping and description of the pinhole method, *Opt. Lett.* 32 (13) (2007) 1908–1910.
 - [31] A. Langar, C. Bouzidi, H. Elhouichet, M. Férid, Er-Yb codoped phosphate glasses with improved gain characteristics for an efficient 1.55 μm broadband optical amplifiers, *J. Lumin.* 2313 (13) (2014) 1–24.
 - [32] W. Ryba-Romanowski, Z. Mazurak, B. Jeżowska-Trzebiatowska, Energy transfer between ytterbium and erbium in phosphorous oxychloride, *J. Mol. Struct.* 46 (1978) 349–354.
 - [33] M. Jaroslaw, K. Krzysztof, M. Zygmunt, Generation investigation of eye-safe microchip lasers pumped by 974 nm and 939 nm wavelength, *Opt. Appl.* 38 (4) (2008) 657–668.
 - [34] X. Zhang, Y. Zhou, J. Ren, D. Lu, H. Yu, Z. Wang, S. Guo, X. Xu, Growth, thermal and laser properties of a new self-frequency-doubling Yb:CNGS crystal, *CrystEngComm* 18 (28) (2016) 5338–5343.
 - [35] A. Jaffrès, B. Viana, P. Loiseau, G. Aka, C. Larat, E. Lallier, Actively Q-Switch Operation of Diode-Pumped Er^{3+} , Yb^{3+} , $\text{Ce}^{3+}:\text{Ca}_2\text{Al}_2\text{SiO}_7$ Single Crystal Laser at 1.5–1.6 μm , *CLEO*, 2013.
 - [36] F. Thibault, D. Pelenc, High Power Diode Pumped Femtosecond Laser Based on Ytterbium Doped YSO and LSO Crystals, *CLEO*, 2005, p. 361.